Project: Sales Dataset Analysis

Objective:

Analyze the dataset containing sales transaction information from a retail business. The goal is to perform comprehensive data analysis, extract meaningful insights, and create interactive visualizations using Microsoft Excel. Additionally, leverage advanced Excel features such as What-If Analysis, Goal Seek, Macros, Power Query, and Power Pivot to deepen your analysis and automate tasks.

Instructions:

1. **Format the Worksheet:**
   o Format the columns for better readability.
   o Use Date formatting for the "Order Date" and "Ship Date" columns to ensure proper sorting and filtering.
   o Apply bold headers and freeze the top row to keep it visible as you scroll through the data.
2. **Clean the Data:**
   o Check for and remove any duplicate rows or blank entries in the dataset.
   o Use data validation to ensure that all data entries (like sales numbers and dates) are consistent and free of errors. For example, ensure Sales and Discount are never negative.
   o Use IF or IFERROR to flag and handle any negative or erroneous sales data that might result from data entry mistakes.
   o Identify and resolve any outliers or incorrect values, especially in Sales and Discount.
3. **Analyze Sales Performance:**
   o Calculate the total sales for each transaction by applying discount on the sales. Create a new column titled "Adjusted Sales" to reflect this. Before doing this step, validate for any negative values.
   o Determine the total revenue, average order value, and total discounts across all transactions.
   o Segment the data by product categories and calculate key metrics (e.g., total adjusted sales, total discounts) for each segment.

4. Use Formulas to Extract Insights:

o Apply COUNTIF/COUNTIFS to count the number of orders for each Product Category or Customer Segment (if available) and group by these categories.

o Calculate the average discount per order using AVERAGEIF based on Customer Segment.

5.Time-Based Analysis:

o Group the data by Date (daily, monthly, quarterly) using PivotTables to assess sales trends over time.

o Identify peak sales periods by analyzing the Adjusted Sales over different time intervals. Look for patterns that could indicate seasonal variations.

o Use Date Functions like MONTH(), YEAR(), WEEKDAY(), and EOMONTH() to categorize and analyze sales performance over different periods.

6. Sales Channel and Product Type Analysis:

o Take Sales Channel related column like "Ship Mode", create a PivotTable to calculate the total Sales, average Sales, and total Discounts for each Sales Channel.

o Identify the best-selling products or product categories based on Quantity and Sales.

7. What-If Analysis & Goal Seek:

o Use Goal Seek to determine the sales target needed to achieve a specific revenue or profit goal. For example, set a goal for Adjusted Sales or Profit and use Goal Seek to find out how many units of a product need to be sold to reach that target.

o Use What-If Analysis to simulate different scenarios such as:

• How would a 10% increase in Sales impact total revenue?

• What would happen if the Discount rate decreased by 5%?

• Create a Scenario Manager to compare various sales and discount scenarios and their effect on Adjusted Sales or Profit.

8. Pivot Tables and Pivot Charts:

o Create PivotTables to aggregate data by different dimensions such as Product Category, Customer Segment, or Ship Mode.

o Use PivotCharts to visualize sales trends and compare performance across different categories (e.g., bar charts for sales by product category, line charts for sales over time).

o Add Slicers to your PivotTables and charts to allow interactive filtering by categories such as Product Category, Ship Mode, or Customer Segment.

9. Create a Sales Dashboard:

• Design a summary dashboard with key performance indicators (KPIs) such as:

o Total sales (Sales or Adjusted Sales)

o Average sales per order

o Total discounts

o Top-performing products or product categories

• Use data validation to allow the user to select specific date ranges or product categories to dynamically update the dashboard.

• Include interactive features such as slicers for Product Category, Ship Mode, or Customer Segment to allow users to filter and view data by specific segments.

11. Automate Tasks Using Macros:

• Record and run macros to automate repetitive tasks, such as:

o Formatting the worksheet (e.g., currency formatting, bolding headers, freezing panes).

o Generating summary tables or charts for Adjusted Sales or Sales by Category.

o Running calculations and formatting pivot tables automatically when new data is added.

• Assign the macros to buttons on the worksheet to make it easy to run them.

12. Power Pivot & Data Modeling:

• Create Two Separate Tables:

• Sheet 2: Products Table Create a table with the following columns:

o Product ID o Product Name o Category o Sub-Category o Sales o Quantity o Discount o Profit

• Sheet 3: Customers Table Create a table with the following columns:

o Customer ID

o Customer Name

o Segment

o Country

o City

o State

o Postal Code

o Region

• Use Power Pivot to create a data model by establishing a relationship between the two tables:

o Connect the Product ID column in the Products Table with the Product ID in the Sales transactions table (or any relevant data that has product-related sales).

o Connect the Customer ID column in the Customers Table with the Customer ID in the Sales transactions table (or any relevant data).

• Create Calculated Columns in Power Pivot:

• Once the tables are connected, create calculated columns in Power Pivot to perform complex calculations, such as Profit Margin (calculated as Profit/Sales).

• Create a Power Pivot-based PivotTable to summarize sales performance using your data model.

13. Insights and Recommendations:

o Summarize the key insights from your analysis, such as:

• Which Sales Channels (e.g., Ship Mode) are driving the highest Sales.

• Which Product Categories are performing best and worst based on Adjusted Sales and Quantity.

- The effect of Discounts on Net Sales and Adjusted Sales.
- Any actionable recommendations for improving Sales Performance, reducing Returns (if applicable), or optimizing Discount strategies.